Saurabh Kashid

Atlanta, GA

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Work Experience

Joulea **Atlanta, GA**

Autonomy Engineer *Feb 2023 – Present*

- Developed a degeneracy detection system in LiDAR SLAM for smooth shifting between Gps and Lidar slam to ensure robust localization.
- Integrated satellite based RTK module (Neo D9S) with GPS (Ublox F9R), enhancing positioning accuracy and reduced electronic interference through optimized shielding.
- Implemented real-time LiDAR obstacle detection on NVIDIA-Orin for safer drone navigation.
- Investigated STD loop closure with GTSAM pose graph optimization techniques to enhance lidar SLAM localization accuracy. .
- Fused LIO, VO, and barometer readings in an EKF for enhanced state estimation.
- Collaborated with contractor on Android app development using DJI SDKs, providing technical guidance on integrating OSDK with MSDK for data transfer.
- Generated building models from point cloud data with Patchwork++ for ground segmentation, optimizing code to achieve a substantial reduction in processing time, enabling faster and more efficient coverage path planning.

PROJECTS

Pb Edge Detection and Texture-Based Segmentation **Jan 2024 – Jan 2024**

- Developed an edge detection pipeline using LM, Gabor, and DoG filter banks combined with χ^2-based gradients on texton, brightness, and color maps.
- ntegrated Canny/Sobel baselines with learned gradients to compute high-quality Pb (probability of boundary) edge maps.

Panoramic Image Stitching Pipeline **Feb 2024 – Feb 2024**

- Trained a HomographyNet from scratch on MS-COCO for direct homography regression in challenging image regions.
- Integrated homographies with feature-matching + RANSAC in an OpenCV pipeline, using Poisson blending for seamless, artifact-free panoramas.

Automatic Camera Calibration **Mar 2024 – Mar 2024**

- Implemented Zhang's method to automatically estimate intrinsic parameters (f_x, f_y, c_x, c_y) and radial distortion (k_1, k_2) from checkerboard images using OpenCV and nonlinear optimization.

6D Pose Detection using Dense Fusion on Linemode Dataset **Mar 2023 – Apr 2023**

- Developed a 6D pose detection system using Dense Fusion on the modified Linemode Dataset.
- Applied state-of-the-art techniques on 6 classes out of 15 of linemode in deep learning and computer vision to improve the performance of the system.

Active Vision **Sep 2022 – Nov 2022**

- Segmented point cloud objects from the major plane using the RANSAC algorithm in ROS2 Gazebo
- Combined point clouds using ICP and estimated grasp points using grasp approximation methods in C++.

EDUCATION

Worcester Polytechnic Institute **Jan 2022 – Dec 2023**

Masters of Science, Robotics Engineering - [GPA - 3.77] *Worcester, MA*

Savitribai Phule Pune University **Aug 2017 – May 2020**

Bachelors of Engineering, Mechanical Engineering - [CGPA - 8.5] *Pune, Maharashtra, India*

TECHNICAL SKILLS

Languages and Technologies: Python, C++, Open3d, NetworkX, , Linux, Docker, Pytorch, OpenCV, Tensorflow, MatLab, PCL, ROS, ROS2, Rviz, Gazebo, Git, GitHub, plotjuggler(ROS), Google$_Ortool, SLAM$

CERTIFICATIONS

- FAA Part 107 Remote Pilot License **31 Mar 2023**